

# Metalearning Deliverables

## Understanding Verification Worksheets

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**Project:** Defending Tracking Control Neural Network  
Against Adversarial Attacks

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62 questions across 7 steps

Each step includes concept checks, worked examples,  
self-assessment checklists, and space for questions.

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## Overview

Seven structured worksheets—one per metalearning step from the background knowledge map. Each contains questions that require you to demonstrate understanding by **explaining in your own words, working through examples by hand, and connecting concepts to the paper**.

These are not quizzes. They are thinking tools. The goal is to surface gaps *before* they block you on a milestone.

## The Seven Steps

| # | Step                            | Key Deliverable It Supports                    |
|---|---------------------------------|------------------------------------------------|
| 1 | Mathematical Foundations        | All milestones (prerequisite)                  |
| 2 | Control Theory Fundamentals     | Feb 2: NN Description                          |
| 3 | Stability Analysis              | Feb 9: Existence/Uniqueness; Feb 16: Stability |
| 4 | Neural Network Concepts         | Mar 2: Block Diagram                           |
| 5 | Adversarial Attacks & Defense   | Apr 6: Simulations (Noise Injection)           |
| 6 | Implementation Skills           | Mar 9: Implementation; Mar 23: Comparison      |
| 7 | Project Timeline & Deliverables | All milestones (planning)                      |

## How to Use

1. **Work through each step in order** ( $1 \rightarrow 7$ ). Earlier steps build foundations for later ones.
2. **Write answers in the provided boxes.**
3. **Be honest with the self-assessment checklists**—unchecked items are study targets.
4. **Use the “Notes / Questions” sections** to capture anything unclear for your next meeting.
5. **Revisit as the semester progresses**—your understanding will deepen.

## Progress Tracker

| Step                               | Started | Completed | Confidence (1–5) |
|------------------------------------|---------|-----------|------------------|
| 1. Mathematical Foundations        | yes     | _____     | _____            |
| 2. Control Theory Fundamentals     | _____   | _____     | _____            |
| 3. Stability Analysis              | _____   | _____     | _____            |
| 4. Neural Network Concepts         | _____   | _____     | _____            |
| 5. Adversarial Attacks & Defense   | _____   | _____     | _____            |
| 6. Implementation Skills           | _____   | _____     | _____            |
| 7. Project Timeline & Deliverables | _____   | _____     | _____            |

# 1 Mathematical Foundations

**Purpose:** Demonstrate understanding of the core mathematics needed to understand the theoretical framework of the tracking control NN.

**When complete:** You can confidently read the paper's equations and trace each step of the derivations.

## 1.1 Difference Equations

### Concept Check

#### Q1: Basic Form & Solutions

Write the general form of a first-order difference equation. Then, given  $x(k+1) = 2x(k) + 1$  with  $x(0) = 0$ , solve for  $x(1)$ ,  $x(2)$ ,  $x(3)$ ,  $x(4)$  by hand.

Your Answer

General Form:  $x_{k+1} = F(x_k)$

Your Answer

Given:  $x(k+1) = 2x(k) + 1$ ,  $x(0) = 0$

$$x(0) = 0 \quad (1)$$

$$x(1) = 2x(0) + 1 = 1, \quad (2)$$

$$x(2) = 2x(1) + 1 = 3 \quad (3)$$

$$x(3) = 2x(2) + 1 = 7 \quad (4)$$

$$x(4) = 2x(3) + 1 = 15 \quad (5)$$

#### Q2: Variable Structure Systems

In your own words, explain what makes a system “variable structure.” Why does the signum function  $\text{sgn}(e)$  create variable structure behavior in the tracking control NN?

Your Answer

A system is of variable structure if the closed loop update law changes discontinuously based on certain conditions of the input i.e. modified by a signum function.

**Q3:** Euler Approximation (Discretization)

Given the continuous-time ODE  $\dot{x} = f(x)$ , derive the Euler approximation that produces the discrete-time form  $x(k+1) = x(k) + \tau f(x(k))$ . What role does  $\tau$  play, and what happens if  $\tau$  is too large?

Your Answer

given  $x' = f(x)$ :

$$x' = f(x)$$

$$x'(t) \approx \frac{x(t+\tau) - x(t)}{\tau} \approx f(x(t))$$

$$x(t+\tau) - x(t) \approx \tau(f(x(t)))$$

$$x(t+\tau) \approx x(t) + \tau(f(x(t)))$$

$$x(k+1) \approx x(k) + \tau(f(x(k))), k = t\tau$$

Tau is the quotient of the current time step over the time horizon (total amount of time steps)

**Q4:** Connecting to the Paper

The paper's core equation is:

$$x(k+1) = x(k) + \tau[f(x(k)) + g(x(k))u(k)]$$

Identify which part comes from the Euler approximation of a continuous-time system, and write out what the original continuous-time ODE would look like.

Your Answer

**1.2 Linear Algebra****Concept Check****Q1:** Matrix Operations

Given

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix},$$

compute  $AB$ ,  $A^\top$ , and  $A^{-1}$ . Show your work.

Your Answer

**Q2: Eigenvalues & Stability**

Compute the eigenvalues of  $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ . For a discrete-time linear system  $x(k+1) = Ax(k)$ , what condition on the eigenvalues guarantees stability? Is this system stable?

Your Answer

**Q3: Diagonal Matrices**

The paper uses  $\beta = \text{diag}(\beta_1, \dots, \beta_n)$  as the learning rate parameter matrix. Explain what a diagonal matrix is, and why using a diagonal matrix for  $\beta$  means each state variable has its own independent learning rate.

Your Answer

## 1.3 Optimization & Gradient

### Concept Check

**Q1: Gradient Descent**

Explain the gradient descent update rule  $\Delta e = -\nabla C(e)$  in plain language. Why does moving in the negative gradient direction minimize the cost?

Your Answer

**Q2: Cost Function Design**

The paper uses  $C(x) = |e|$  where  $e$  is the tracking error. Why is absolute value a valid cost function? What property does it have at  $e = 0$ ? How does this connect to the signum function (hint:  $\frac{d|e|}{de} = \text{sgn}(e)$ )?

Your Answer

**Q3: Convexity**

Define what it means for a cost function to be convex. Why does convexity guarantee a unique global minimum? Is  $C(x) = |e|$  convex?

Your Answer

### Self-Assessment Checklist

- ☐ I can solve difference equations iteratively given initial conditions
- ☐ I can explain why the signum function creates variable structure
- ☐ I can derive the Euler approximation from a continuous ODE
- ☐ I can perform basic matrix operations (multiply, transpose, inverse)
- ☐ I can compute eigenvalues and relate them to system stability
- ☐ I can explain gradient descent and why the cost function  $C = |e|$  is chosen
- ☐ I can connect all of the above to the paper's system model equation

Notes / Questions for Professor Tymoshchuk

## 2 Control Theory Fundamentals

**Purpose:** Demonstrate understanding of discrete-time dynamics, affine nonlinear systems, tracking control, and optimal control via the Hamilton–Jacobi–Bellman equation.

**When complete:** You can explain the full control pipeline from system model to optimal control law, and why each design choice was made.

### 2.1 Discrete-Time Dynamic Systems

#### Concept Check

##### Q1: State-Space Representation

Write the paper’s system model

$$x(k+1) = x(k) + \tau[f(x(k)) + g(x(k))u(k)]$$

and identify each term:

- What is  $x(k) \in \mathbb{R}^n$  and what does it represent physically?
- What is  $u(k) \in \mathbb{R}^m$  and what does it represent?
- What is the difference between  $f(x)$  and  $g(x)$ ?
- What is  $\tau$ ?

Your Answer

##### Q2: Time Step Selection

The paper requires  $0 < \tau < 2|x(k)|/h(x(k))$  for stability. Explain in your own words why  $\tau$  cannot be arbitrary. What happens physically if  $\tau$  is too large? What if  $\tau$  is too small?

Your Answer

##### Q3: Worked Example

Consider a scalar system ( $n = 1$ ) with  $f(x) = -x$ ,  $g(x) = 1$ ,  $\tau = 0.1$ . Starting from  $x(0) = 5$  with constant input  $u(k) = 0$ , compute  $x(1)$ ,  $x(2)$ ,  $x(3)$ . Is the state converging?

Your Answer

## 2.2 Affine Nonlinear Systems

### Concept Check

#### Q1: Affine-in-the-Input

The continuous-time form is  $\dot{x} = f(x) + g(x)u$ . Explain precisely what “affine in the input” means. Why is the control input  $u$  appearing *linearly* important for solving the control problem, even though  $f(x)$  and  $g(x)$  can be arbitrarily nonlinear?

Your Answer

#### Q2: Affine vs. Linear

Distinguish between:

- A **linear** system:  $\dot{x} = Ax + Bu$
- An **affine** system:  $\dot{x} = f(x) + g(x)u$
- A general **nonlinear** system:  $\dot{x} = F(x, u)$

Why can't the paper's methods apply to the general nonlinear case?

Your Answer

#### Q3: Controllability

The paper requires  $g(x) \neq 0$ . Explain what would happen if  $g(x) = 0$  for some state  $x$ . Why would the system be “uncontrollable” at that point?

Your Answer

## 2.3 Tracking Control Problem

### Concept Check

#### Q1: Tracking Error

Define the tracking error  $e(k) = x(k) - r(k)$ . In the ACC (Adaptive Cruise Control) context, what does  $x(k)$  represent and what does  $r(k)$  represent? What does  $e(k) = 0$  mean physically?

Your Answer

**Q2: Control Law Derivation**

The paper derives the control law:

$$u(k) = g^{-1}(x(k))[-\beta \operatorname{sgn}(x(k) - r(k)) + \Delta r(k) - f(x(k))]$$

Break this equation apart term by term:

- What does  $g^{-1}(x(k))$  do?
- What does  $-\beta \operatorname{sgn}(x(k) - r(k))$  do?
- What does  $\Delta r(k)$  contribute?
- What does  $-f(x(k))$  accomplish?

Your Answer

**Q3: Closed-Loop Tracking Equation**

Substituting the control law into the system model yields the tracking equation:

$$x(k+1) = x(k) - \beta \operatorname{sgn}(x(k) - r(k)) + \Delta r(k)$$

Verify this by substituting  $u(k)$  into  $x(k+1) = x(k) + \tau[f(x(k)) + g(x(k))u(k)]$  step by step.

Your Answer

## 2.4 Optimal Control & HJB Equation

### Concept Check

**Q1: Value Function**

Define the value function  $V(x)$  in optimal control. What does it represent? Why is it useful?

Your Answer

**Q2: HJB Equation**

Write out the Hamilton–Jacobi–Bellman equation:

$$Q(x) + V_x^\top f(x) - \frac{1}{4} V_x^\top g(x) R^{-1} g^\top(x) V_x = 0$$

Identify the role of each term:

- $Q(x)$ : ?
- $V_x^\top f(x)$ : ?
- The quadratic term: ?

Your Answer

**Q3:** Analytical vs. Numerical

The paper claims to derive an *analytical* solution to the HJB equation. Why is this significant? What do most other approaches do instead, and what are the downsides?

Your Answer

**Self-Assessment Checklist**

- ☐ I can write and interpret the discrete-time system model equation
- ☐ I can explain why the system is “affine in the input” and why that matters
- ☐ I can define tracking error and explain the tracking control objective
- ☐ I can break down the control law  $u(k)$  term by term
- ☐ I can derive the closed-loop tracking equation by substitution
- ☐ I can state the HJB equation and explain each term’s role
- ☐ I can explain why an analytical HJB solution is a significant contribution

Notes / Questions for Professor Tymoshchuk

## 3 Stability Analysis

**Purpose:** Demonstrate understanding of Lyapunov stability, existence/uniqueness of steady states, finite-time convergence, and sliding mode analysis—the theoretical backbone proving the defense mechanism works.

**When complete:** You can independently walk through the paper’s stability proofs and explain why the defended tracking control NN is guaranteed to work correctly.

### 3.1 Lyapunov Stability Theory

#### Concept Check

##### Q1: Lyapunov’s Second Method

State the two conditions required for Lyapunov stability:

1. Condition on  $V(x)$ : ?
2. Condition on  $\Delta V(k) = V(k+1) - V(k)$ : ?

In your own words, explain the intuition: why do these two conditions guarantee stability?

Your Answer

##### Q2: Lyapunov Function Construction

The paper uses  $W(k) = |x(k)|$  as the Lyapunov candidate function. Verify that it satisfies the first condition (positive definite): Is  $W(x) > 0$  for  $x \neq 0$  and  $W(0) = 0$ ?

Your Answer

##### Q3: Global Asymptotic Stability

Define global asymptotic stability. What is the difference between:

- Stability (trajectories stay close)
- Asymptotic stability (trajectories converge)
- Global asymptotic stability (converge from *any* initial condition)

Why is *global* stability important for this project’s defense guarantees?

Your Answer

##### Q4: Radial Unboundedness

What does  $V(x) \rightarrow \infty$  as  $\|x\| \rightarrow \infty$  mean? Why is this condition needed for *global* (not just local) stability?

Your Answer

### 3.2 Existence & Uniqueness of Steady States

#### Concept Check

##### Q1: Equilibrium Analysis

At a steady state,  $x(k+1) = x(k)$ , so the change is zero. Write the equilibrium condition from the tracking equation:

$$\frac{x(k+1) - x(k)}{\tau} = h(x) S(x) = 0$$

What are the two ways this product can equal zero? Which corresponds to the correct steady state?

Your Answer

##### Q2: Why Uniqueness Matters

Suppose the system had multiple steady states. Explain in your own words why this would be problematic for:

- (a) Normal operation (tracking a reference)
- (b) Operation under adversarial attack

Your Answer

##### Q3: Connecting to Defense

If an adversary perturbs the system, the steady state analysis guarantees the system will still converge to the *correct* steady state (not a false one). Explain why having a unique steady state is the foundation of the defense mechanism.

Your Answer

### 3.3 Finite-Time Convergence

#### Concept Check

**Q1: Lemma 2 Statement**

State Lemma 2 in your own words: trajectories converge to a  $\delta$ -neighborhood of the steady state in at most  $m$  steps, where  $m \leq (x(1) - \delta)/\sigma + 1$  and  $\sigma = \tau h(\delta)$ .

What do each of these quantities represent?

- $\delta$ : ?
- $\sigma$ : ?
- $m$ : ?

Your Answer

**Q2: Convergence Bound Derivation**

The proof uses  $W(x(m)) \leq W(x(1)) - \sigma(m - 1)$ . Explain the logic:

- Why does  $W$  decrease by at least  $\sigma$  each step?
- How does setting  $W(x(m)) \geq 0$  give you the upper bound on  $m$ ?

Your Answer

**Q3: Invariant Set Property**

Once the trajectory enters the  $\delta$ -neighborhood, it stays there (the set  $\Phi$  is invariant). Why is this property important? What would happen without it?

Your Answer

### 3.4 Sliding Mode Analysis

#### Concept Check

**Q1: Sliding Mode Conditions**

State the two sliding mode conditions:

- $\Delta x(k) > 0$  when  $x(k) \rightarrow -0$
- $\Delta x(k) < 0$  when  $x(k) \rightarrow +0$

Draw a simple diagram or describe in words what this means: the trajectory oscillates around  $x = 0$ .

Your Answer

**Q2: Chattering Phenomenon**

What is chattering? Why does it occur in discrete-time systems with signum switching? Is chattering a problem or a feature in this context?

Your Answer

**Q3: Full Stability Story**

Put it all together. Trace the complete stability narrative:

1. System starts at arbitrary initial condition  $x(0)$
- 2–5. *What happens next? Fill in each phase until steady state.*

Your Answer

**Self-Assessment Checklist**

- ☐ I can state Lyapunov's second method and verify a candidate function
- ☐ I can explain global asymptotic stability and radial unboundedness
- ☐ I can derive the equilibrium condition and explain uniqueness
- ☐ I can state the finite-time convergence bound and explain the proof logic
- ☐ I can describe sliding mode behavior and chattering
- ☐ I can tell the complete stability story from initial condition to steady state

Notes / Questions for Professor Tymoshchuk

## 4 Neural Network Concepts

**Purpose:** Demonstrate understanding of the NN structure used for tracking control. This is **not** a deep learning NN—it is a control-theoretic neural network with specific architectural components.

**When complete:** You can describe every component of the functional block diagram and explain why the architecture is called a “neural network.”

### 4.1 Basic NN Architecture

#### Concept Check

##### Q1: Not Deep Learning

This tracking control NN is fundamentally different from a deep learning neural network. Explain:

- In what sense is it called a “neural network”?
- What plays the role of activation functions?
- What plays the role of weights/learning?
- What acts as the “teacher” signal (supervised learning interpretation)?

Your Answer

##### Q2: Signum Activation Function

Write the definition of the signum function:

$$\text{sgn}(e) = \begin{cases} ? & \text{if } e > 0 \\ ? & \text{if } e = 0 \\ ? & \text{if } e < 0 \end{cases}$$

How does this create “variable structure” behavior? Compare it to smooth activation functions (sigmoid, ReLU) used in deep learning.

Your Answer

##### Q3: Learning Rate Parameter $\beta$

The parameter  $\beta$  controls convergence speed. Explain:

- What happens if  $\beta$  is too large?
- What happens if  $\beta$  is too small?
- How is this analogous to learning rate in gradient descent?

Your Answer

**Q4: Supervised Learning Interpretation**

The reference trajectory  $r(k)$  acts as the “teacher signal.” Explain how the tracking control loop mirrors supervised learning:

- Training data = ?
- Prediction = ?
- Error signal = ?
- Update rule = ?

Your Answer

**4.2 Functional Block Diagram (Fig. 1)****Concept Check****Q1: Component Inventory**

The block diagram contains:

- $3n$  adders
- $n$  controlled switches (signum)
- $n$  amplifiers ( $\beta$ )
- $n$  digital differentiators ( $\Delta$ )
- $n$  blocks for  $f(x)$
- $n \times m$  blocks for  $g(x)$
- $n$  NN solvers (LAESS)

For a system with  $n = 2$  states and  $m = 1$  input, calculate the total number of each component.

Your Answer

**Q2: Key Components**

Match each component to its mathematical function:

| Component                   | Symbol | What it computes |
|-----------------------------|--------|------------------|
| Adder (+)                   |        |                  |
| Sign ( $S$ )                |        |                  |
| Amplifier ( $\beta$ )       |        |                  |
| Differentiator ( $\Delta$ ) |        |                  |

Your Answer

**Q3: LAESS—Linear Algebraic Equation Solver**

The LAESS solves for  $u(k)$  from the control equation. Explain:

- What equation does it solve?
- Why can it be implemented as a sub-network?
- What inputs does it need and what output does it produce?

Your Answer

**Q4: Signal Flow**

Trace the signal flow through the block diagram for one time step:

1. Input: current state  $x(k)$  and reference  $r(k)$
- 2– $N$ . Fill in each processing step.
- $N+1$ . Output: next state  $x(k+1)$

Your Answer

**Q5: Draw Your Own**

In the space below, sketch or describe a simplified version of the block diagram for  $n = 1$ ,  $m = 1$ . Label every block and connection.

Your Answer

**Self-Assessment Checklist**

- ☐ I can explain why this is called a “neural network” despite not being deep learning
- ☐ I can define the signum activation function and explain its variable structure effect
- ☐ I can explain the role of  $\beta$  as a learning rate parameter

- ☐ I can list all components in the functional block diagram
- ☐ I can trace signal flow through the block diagram for one time step
- ☐ I can explain the LAESS sub-network and its purpose
- ☐ I can draw a simplified block diagram for a scalar system

Notes / Questions for Professor Tymoshchuk

## 5 Adversarial Attacks & Defense

**Purpose:** Demonstrate understanding of threats to control systems and how the tracking control NN defends against them. This is the core novelty of the project.

**When complete:** You can explain what adversarial attacks do to a control system, why this NN architecture is inherently robust, and how the defense connects to the stability proofs.

### 5.1 Types of Adversarial Attacks

#### Concept Check

##### Q1: Spoofing Attacks

Define a spoofing attack in the context of a control system. Give a concrete example: if the tracking control NN is controlling vehicle speed (ACC), what would a spoofing attack look like?

Your Answer

##### Q2: Denial of Service (DoS)

Define a DoS attack in the control context. How does it differ from spoofing? What is the effect on the controller if legitimate sensor data is blocked?

Your Answer

##### Q3: Attack Modeling

In mathematical terms, how does the paper model an adversarial attack? Is it:

- (a) Additive noise on the state:  $x_{\text{measured}}(k) = x(k) + \delta(k)$
- (b) Corruption of the control input:  $u_{\text{actual}}(k) = u(k) + \delta(k)$
- (c) Something else?

Explain how the attack enters the system equations.

Your Answer

### 5.2 Defense Mechanisms

#### Concept Check

##### Q1: Robustness to Disturbances

The paper shows the controller works correctly if nonlinearity disturbances are bounded:

- Signum disturbance:  $0 \leq \rho < 1$
- Function  $h$  disturbance: bounded by  $\sqrt{f^2 + R^{-1}Qg^2}$

In your own words, explain what “bounded disturbance” means and why there are limits to what the defense can handle.

Your Answer

## Q2: Why This Architecture Defends

The key claim: the tracking control NN maintains correct tracking **despite** adversarial perturbations, as long as they are bounded. Trace the logic:

1. Attack perturbs the system state
2. Stability proof guarantees... *(fill in)*
3. Uniqueness of steady state means... *(fill in)*
4. Finite-time convergence ensures... *(fill in)*

Your Answer

## Q3: Estimation-Based Defense

The project mentions using Kalman filters or observers to estimate the true state. Explain the basic idea: if the attacker corrupts the sensor reading, how can you estimate the actual state?

Your Answer

## Q4: Integration with IDS

An Intrusion Detection System (IDS) can trigger defense mechanisms. Describe how an IDS might work alongside the tracking control NN:

- What does the IDS detect?
- What action does it trigger?
- How does this complement the NN’s inherent robustness?

Your Answer

## 5.3 CAN Bus & Vehicle Networks

## Concept Check

### Q1: CAN Protocol Basics

Explain the basic properties of the CAN (Controller Area Network) bus:

- How do ECUs communicate?
- Why is the broadcast nature of CAN a vulnerability?
- What authentication/encryption does CAN have?

Your Answer

### Q2: Motivating the Defense

Connect CAN bus vulnerabilities to the need for defended tracking control:

- ECUs communicate sensor data (speed, distance) over CAN
- An attacker on the CAN bus can... *(fill in)*
- This means the tracking controller receives... *(fill in)*
- The defense mechanism ensures... *(fill in)*

Your Answer

## 5.4 The Big Picture

### Synthesis Question

#### Q1: End-to-End Defense Story

Write a one-paragraph narrative that connects all of Step 5: Start with the threat (CAN bus vulnerability), describe the attack type, explain how it affects the control system, and show how the tracking control NN's theoretical properties (stability, uniqueness, convergence) provide defense.

Your Answer

### Self-Assessment Checklist

- ☐ I can describe spoofing and DoS attacks in a control system context
- ☐ I can explain how attacks are modeled mathematically in the paper
- ☐ I can state the robustness bounds (when the defense works and when it fails)

- ☐ I can trace the defense logic through stability  $\rightarrow$  uniqueness  $\rightarrow$  convergence
- ☐ I can explain CAN bus vulnerabilities and why they motivate this research
- ☐ I can tell the end-to-end defense story in one coherent narrative

Notes / Questions for Professor Tymoshchuk

## 6 Implementation Skills

**Purpose:** Demonstrate readiness to implement the tracking control NN in MATLAB, Python, and C++/CUDA. Not about mastery of each language, but about knowing the right tools and libraries for control system simulation.

**When complete:** You can set up a simulation environment in each language and implement basic control system operations.

### 6.1 MATLAB Implementation

#### Concept Check

##### Q1: Core Functions

For each MATLAB function, describe what it does and when you would use it in this project:

| Function            | Purpose | When to Use |
|---------------------|---------|-------------|
| <code>ss()</code>   |         |             |
| <code>tf()</code>   |         |             |
| <code>step()</code> |         |             |
| <code>bode()</code> |         |             |

Your Answer

##### Q2: Simulink

What is Simulink and how does it relate to the functional block diagram from Step 4? Would you implement the block diagram directly in Simulink or in MATLAB script? What are the trade-offs?

Your Answer

##### Q3: MATLAB Simulation Skeleton

Write pseudocode (or actual MATLAB code) for the basic simulation loop of the tracking control NN:

- Initialize state  $x(0)$ , reference trajectory  $r(k)$ , parameters  $(\tau, \beta, f, g)$
- Loop over  $k = 0$  to  $N$ : compute  $u(k)$ , update  $x(k+1)$ , store results
- Plot  $x(k)$  vs.  $r(k)$

Code / Pseudocode

## 6.2 Python Implementation

### Concept Check

#### Q1: Library Mapping

Map MATLAB functionality to Python libraries:

| MATLAB                  | Python Equivalent | Library |
|-------------------------|-------------------|---------|
| Matrix operations       |                   |         |
| Control System Toolbox  |                   |         |
| ODE / system simulation |                   |         |
| Plotting                |                   |         |

Your Answer

#### Q2: NumPy for the System Model

Write Python code (using NumPy) to compute one time step of the system model:

$$x(k+1) = x(k) + \tau(f(x(k)) + g(x(k))u(k))$$

Assume  $\mathbf{x}$  is an  $n$ -dimensional array,  $\mathbf{u}$  is an  $m$ -dimensional array, and  $\mathbf{f}()$ ,  $\mathbf{g}()$  are functions.

Code / Pseudocode

#### Q3: python-control Library

What does the `python-control` library provide? How would you use it for this project? Give an example function call.

Your Answer

## 6.3 C++ and CUDA

### Concept Check

#### Q1: Why C++?

Why would you implement the tracking control NN in C++ when MATLAB and Python are easier? What practical applications require C++ performance?

Your Answer

#### Q2: Eigen Library

The Eigen library provides matrix operations in C++. Write pseudocode showing how you would represent the state vector  $x(k)$  and compute one system update step using Eigen-style operations.

Code / Pseudocode

#### Q3: CUDA Parallelism

For the block diagram with  $n$  state variables:

- Which computations can be parallelized across states?
- Which computations must be sequential?
- How would you map the  $n$  blocks of  $f(x)$  and  $n \times m$  blocks of  $g(x)$  to GPU threads?

Your Answer

#### Q4: CUDA Necessity

Under what conditions would CUDA actually provide a speedup over CPU computation for this problem? (Hint: think about the size of  $n$  and  $m$ , and the number of simulation steps.)

Your Answer

## 6.4 Implementation Comparison Framework

### Synthesis Question

#### Q1: Comparison Criteria

The project requires comparing implementations across languages. Fill in the table:

| Criterion             | How to Measure | Expected Winner |
|-----------------------|----------------|-----------------|
| Execution speed       |                |                 |
| Code complexity       |                |                 |
| Numerical accuracy    |                |                 |
| Ease of visualization |                |                 |
| Real-time capability  |                |                 |

Your Answer

### Self-Assessment Checklist

- ☐ I can identify the MATLAB functions needed for control system simulation
- ☐ I can write a basic simulation loop in MATLAB pseudocode
- ☐ I can map MATLAB tools to Python equivalents (NumPy, SciPy, `python-control`)
- ☐ I can implement one time step of the system model in Python/NumPy
- ☐ I can explain why C++ and CUDA matter for real-time control
- ☐ I can identify which computations parallelize on GPU
- ☐ I can define comparison criteria for cross-language implementation evaluation

Notes / Questions for Professor Tymoshchuk

## 7 Project Timeline & Deliverables

**Purpose:** Demonstrate understanding of the overall research plan, how the metalearning steps map to milestones, and what each deliverable requires.

**When complete:** You can explain how every week's work builds on previous weeks, and you know exactly what "done" looks like for each milestone.

### 7.1 Milestone Mapping

#### Concept Check

##### Q1: Milestone-to-Metalearning Map

For each milestone, identify which metalearning steps (1–6) provide the background knowledge needed:

| Milestone                          | Due    | Required Steps | Why |
|------------------------------------|--------|----------------|-----|
| Description of tracking control NN | Feb 2  |                |     |
| Existence & uniqueness             | Feb 9  |                |     |
| Stability & convergence            | Feb 16 |                |     |
| Functional block diagram           | Mar 2  |                |     |
| Implementation                     | Mar 9  |                |     |
| Midterm report                     | Mar 16 |                |     |
| Compare implementations            | Mar 23 |                |     |
| Compare with analogs               | Mar 30 |                |     |
| Simulations (noise injection)      | Apr 6  |                |     |
| Final report & poster              | Apr 10 |                |     |

Your Answer

### 7.2 Deliverable Definitions

#### Concept Check

##### Q1: "Description of Tracking Control NN" (Feb 2)—COMPLETED

Describe what this deliverable contains. What equations, variables, and parameters must be presented? What does "by the difference equation with variable structure" mean concretely?

Your Answer

**Q2:** “Existence & Uniqueness Analysis” (Feb 9)—IN PROGRESS

What specific claims must this deliverable prove or demonstrate?

- Existence of steady states means: ?
- Uniqueness means: ?
- “In the presence of adversarial attacks” adds what constraint?

What format should this take (proof sketch, narrative explanation, mathematical derivation)?

Your Answer

**Q3:** “Stability & Convergence Analysis” (Feb 16)

What are the two separate claims in this deliverable?

1. “Globally stable” means: ?
2. “Globally convergent in a finite number of steps” means: ?

How does Lemma 2 provide the convergence bound?

Your Answer

**Q4:** “Functional Block Diagram” (Mar 2)

What should the block diagram show?

- All components from Step 4.2
- Signal flow for: ?
- The defending part specifically includes: ?

What tool will you use to create it?

Your Answer

**Q5:** “Compare with Analogs” (Mar 30)

What are “competitive analogs”? What dimensions of comparison does the research plan specify?

- Computational complexity: ?
- Hardware implementation complexity: ?
- What other analogs exist in the literature (Refs. [1]–[3] of the research plan)?

Your Answer

**Q6:** “Simulations—Noise Injection” (Apr 6)

This is the project’s novelty. Describe:

- What noise injection means in this context
- What the simulations should demonstrate
- What plots/figures you expect to produce
- How you will compare simulation results with theoretical predictions

Your Answer

## 7.3 Risk Assessment

### Concept Check

**Q1:** Dependencies & Bottlenecks

Identify which milestones are most at risk:

- Which milestone depends most heavily on understanding from previous milestones?
- Which milestone has the most technical uncertainty?
- What is your backup plan if CUDA implementation proves too complex?

Your Answer

**Q2:** Weekly Planning

For the current week (week of Feb 10), write your specific plan:

- What must be completed?
- What questions need to be answered?
- What will you read/study?
- What will you write/code?

Your Answer

## 7.4 Research Communication

### Concept Check

#### Q1: Student Log / Research Journal

The CAHSI program requires a student log. For each weekly entry, what should you include?

- Plan language (verbatim from research plan): ?
- Your own explanation (avoiding plagiarism): ?
- Questions-forward draft: ?

Your Answer

#### Q2: Elevator Pitch

In 3–4 sentences, explain the entire research project to someone with a CS background but no control theory knowledge. Cover: the problem, the approach, the novelty, and the expected outcome.

Your Answer

### Self-Assessment Checklist

- ☐ I can map each milestone to its required background knowledge
- ☐ I can define what “done” looks like for each deliverable
- ☐ I understand the existence/uniqueness deliverable requirements (current focus)
- ☐ I know what the functional block diagram must contain
- ☐ I can explain the noise injection simulation plan
- ☐ I can identify project risks and dependencies
- ☐ I can explain the entire project in a 30-second elevator pitch

Notes / Questions for Professor Tymoshchuk